

GreenDev AI - Sustainability Report

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1. Code Metrics

Functions	0
Loops	1
Nested Loops	0
Lines of Code	49
Complexity	6
Task Type	io-heavy
Recursion	False

2. Energy Measurement (CodeCarbon)

Energy Consumed	0.000010 kWh
CO2 Emitted	0.0036 g
Execution Time	0.0060 s
Measurement Mode	TDP-estimation (Fallback)

3. Real RAPL Benchmark (Energy-Languages Dataset)

Language	Energy (J)	Time (s)	Memory (MB)
Python	5200.3000	28.9000	68.4000
C	78.4000	1.9000	3.2000
C++	85.2000	2.1000	3.6000
Java	462.8000	5.4000	280.0000

4. SCI Scores

Estimated SCI	0.0036 gCO2eq/run
Real SCI	0.7886 gCO2eq/run
Deviation	99.54%
Anomaly	True

5. Green Score

Overall	84 / 100
Performance	91
Energy	82
Carbon	83.1703871
Maintainability	81.0

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6. Carbon Cost Projection

Per Run	0.0036 g CO2
Daily Runs (assumed)	150
Yearly CO2 (current)	0.2000 kg
Yearly CO2 (optimized)	0.1600 kg
Potential Savings	20%

Planner Reflection Note

Telemetry flagged: SCI deviation is 99.5%, exceeding the $\pm 15\%$ hardware benchmark variance limit.